

What Symmetry Buys a Learned Motion Planner

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Abstract—Learning-based motion planners pay at training what classical planners pay per query. Trained in world coordinates, they relearn the same motion at every position and orientation. Existing work restores the missing rigid-body equivariance in the training data, in the inference operator, or in the weights, and each carries a cost. We ask how much of that equivariance the planning query supplies for free. A start s and a goal g determine a frame in closed form, with origin at their midpoint and first axis along $g - s$. Expressing trajectory and obstacles in that frame removes three translations and two rotations of $SE(3)$, at initialisation, for one cross product per query and with no constraint on the architecture. A single rotation about the start-goal axis remains, and no rule that reads only the query removes it. On a cluttered 3D benchmark, holding architecture, data and budget fixed, the frame raises the held-out collision-free rate from 14.60% to 51.10%, where a straight segment from start to goal scores 15.6% and the world-frame model does not beat it. We build all three mechanisms for the residual rotation and each is worth under a point, though the equivariant backbone reaches any given level two to three times sooner. What the representation supplies therefore dominates what any mechanism enforces, and the standard diagnostic does not see the difference: two models with indistinguishable non-equivariance residuals differ by 28 points. Calibrated against a non-symmetry intervention, the frame is not even the largest effect available, since local geometry is worth +40.0 where the frame is worth +36.5.

I. INTRODUCTION

Motion planning is a fundamental problem in robotics, requiring the generation of collision-free, kinematically feasible trajectories from a start to a goal configuration in complex environments. Equivariance to rigid-body transformations is a cornerstone of generalization in this setting. It ensures that a planner solves structurally identical tasks identically, regardless of where in the world they are posed. Sampling-based [13] and optimization-based [22], [27] algorithms are equivariant by construction, since they act on the geometry directly, and they offer completeness or convergence guarantees. Their per-query computational cost limits real-time applicability. Learning-based planners [2], [8], [11], [20] pay at training what those planners pay per query, and have emerged as a promising alternative. They do not inherit the symmetry. A network trained on world coordinates relearns the same motion at every position and orientation.

Existing work restores it in one of three parts of the pipeline. In the data, through augmentation. In the inference operator, through frame averaging or learned canonicalization [18], [12]. Or in the weights, through architectures equivariant by construction [4], [5], [9], [23], as adopted in recent robot learning systems [21], [24]. Each buys the symmetry at a price: a larger training distribution, additional forward passes

per query, or a restricted hypothesis class. Before paying any of them, it is worth asking how much of the symmetry the problem statement already supplies.

A planning query names a start and a goal, and those two points determine a canonical frame. One can place the origin at their midpoint, align the first axis with the direction between them, and express every waypoint and every obstacle in that frame. Three translational and two rotational degrees of freedom vanish exactly, at initialisation, at the cost of one cross product per query and with no constraint on the architecture. The six numbers that conditioned the network on the query collapse to one invariant scalar. The frame needs a second axis to be determined, and ours takes it from a fixed world axis, so rotating the whole problem rolls the frame rather than leaving it unchanged. What survives is exactly that rotation: $SE(3)$ collapses to $SO(2)$. No rule that reads only the query can fix the remainder, and a rule that also reads the scene cannot fix it continuously in general [6], so the sixth degree of freedom is left to one of the three mechanisms above, and we build all three. Fig. 1 takes the group apart one stage at a time.

We build all three on a cluttered 3D benchmark, hold the architecture, data and training budget fixed, and score every model on problems it never saw, alongside a straight line from start to goal and three classical planners. The frame is worth +36.5 points of held-out collision-free rate; the three mechanisms for the sixth are worth +0.8, +0.68 and +0.15. That line scores 15.6%, which the world-frame model does not reach.

Our contributions are as follows.

- **Five of six degrees of freedom are free.** The start and the goal determine a canonical frame, and expressing the task in it removes five degrees of freedom exactly, at initialisation, for one cross product per query and with no change to the architecture, the data or the budget. It raises the held-out collision-free rate from 14.60% to 51.10%, against 15.6% for a straight segment from start to goal, which the world-frame model does not beat. On an L-shaped rigid body, which has an orientation as well as a position, the reduced model reaches $3.2\times$ the world-frame model at comparable best-validation checkpoints. Replacing flow matching with DDPM, at the same number of forward passes per sample, moves the reduced model to 50.52% against 13.51% for the world frame, so the effect does not belong to the sampler.
- **All three mechanisms for the sixth, measured against each other.** Frame averaging and the $SO(2)$ -equivariant backbone were trained to convergence, and neither is

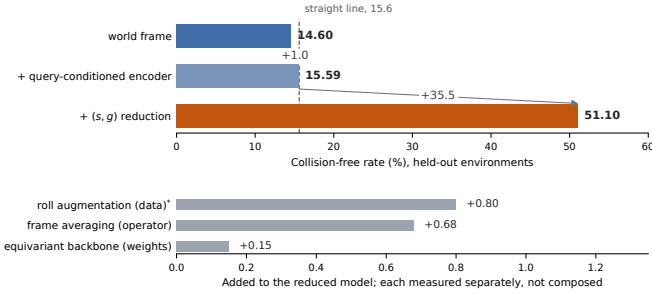


Fig. 1. **Sixty environments, trained to convergence.** *Top:* the frame determined by the start and the goal is worth +35.5 points over the query-conditioned encoder and +36.5 over the plain world frame. Showing the obstacle encoder the same query, with no change of frame, is worth +1.0. The dashed line is a straight segment from start to goal on the identical problems, which the world-frame model does not beat. *Bottom:* the three mechanisms for the one degree of freedom the frame cannot remove. Each is measured against its own baseline. The three are alternatives rather than a decomposition, so they are not composed here and were never composed in the experiments. Two of them are converged measurements. Roll augmentation, starred, is the 20-epoch grid, where its effect was already not significant across three seeds and was not run further.

worth more than a point. Roll augmentation was measured only at a fixed budget, where its +0.8 already sat inside the spread across three seeds. The $SO(2)$ -equivariant backbone reaches any given level two to three times sooner, so what the symmetry buys is optimisation rather than accuracy.

- **A bound on what post-hoc symmetrisation can remove, and its limits as a predictor.** We measure the non-equivariance residual r of the trained field and prove that no symmetrisation of that field can remove more than $r^2 \|\mathcal{A}f\|^2$ of squared error, and that frame averaging reaches this bound. The reduced model has $r = 0.017$. That bound is under 3×10^{-4} of the field’s squared magnitude, consistent with frame averaging being worth +0.68. The residual does not predict what a training-time mechanism is worth: two models with indistinguishable r differ by 28 points.
- **Calibration against a non-symmetry intervention.** Supplying each waypoint with its distance and direction to the nearest obstacle is worth +40.0 points where the frame is worth +36.5, so the representation change studied here is not the largest effect available on this benchmark. With local geometry supplied the frame still adds +24.8, and the two together reach 79.37, below the 91.1 that adding the two effects would predict.

II. RELATED WORK

Classical planners. Sampling-based [13] and optimization-based [22], [27] planners act directly on geometry, so a rigid transformation of the problem transforms the solution. They offer probabilistic completeness or local convergence, at a per-query cost that follows from searching or optimizing at inference time. We use RRT-Connect and CHOMP to generate

expert trajectories, and all three to bound what is achievable on our benchmark.

Learning-based planners. MPNet [20] and Motion Policy Networks [8] move planning into a network evaluated in milliseconds. Diffuser [11], Motion Planning Diffusion [2] and Diffusion Policy [3] generate whole trajectories with a denoising model, and flow matching [14] replaces the denoising chain with a learned velocity field, which is the objective we use. None of them inherits the symmetry that the classical planners get from the geometry for free.

Restoring the symmetry. The general framing is geometric, and equivariant models are known to carry a strict generalisation benefit under the right conditions [7]. Group-equivariant convolution [4], tensor field networks [23], $SE(3)$ -Transformers [9] and vector neurons [5] build the constraint into the weights, and robot learning has adopted them widely [21], [24], including in diffusion planners [1]. Frame averaging [18] and learned canonicalization [12] act at inference instead, and the projection property our bound rests on is theirs. Continuous canonicalization is impossible in general, which motivates weighted frames [6]. The Lie derivative [10] measures how equivariant a trained model became. That is the quantity we bound, and we find it is not sufficient to predict what a mechanism is worth. Frames read off the start and the goal are common practice in manipulation policy learning, and agent- and query-centric frames are a standard design choice in trajectory forecasting, where invariant and non-invariant encoders have been compared [25], [26]. We do not claim the construction. We claim what it is worth in motion planning, against a floor and against the three alternatives.

III. METHOD

In this section, we begin by defining the generative planning objective and the symmetry the task carries (Sec. III-A). Sec. III-B then splits $SE(3)$ into the part a query fixes and the part it leaves behind, and identifies the remainder as $SO(2)$. That remainder does not yield to the same treatment: Sec. III-C shows that no rule reading only the query removes it, that every such rule is discontinuous, and that a scene with a symmetry of its own leaves even a scene-reading rule ambiguous. Finally, Sec. III-D sets out the three places a symmetry can be imposed, in the training data, in the inference operator, and in the weights, and defines the non-equivariance residual that bounds what the operator can remove.

A. Problem formulation

We formulate motion planning in the workspace $\mathbb{R}^3$. A task is the tuple $\mathcal{P} = (s, g, \mathcal{O})$, where $s, g \in \mathcal{C}$ are the start and goal configurations, $\mathcal{C} = \mathbb{R}^3$ for the point mass studied throughout and $\mathcal{C} = SE(3)$ for the rigid-body domain of Sec. V-D, and $\mathcal{O}$ is a set of convex obstacles inducing a signed distance field $d_{\mathcal{O}} : \mathbb{R}^3 \rightarrow \mathbb{R}$. The robot is a sphere of radius ρ centered at p . For the rigid body of Sec. V-D it is a union of five such spheres rigidly attached to the pose, at centres $c_j(p)$; the point mass is the single-sphere case $c_1(p) = p$. A trajectory is a waypoint sequence $x = [p_1, \dots, p_N] \in \mathcal{C}^N$ and $\gamma(x)$ the path it traces,

piecewise linear in position and geodesic in orientation. The feasible set of $\mathcal{P}$ is

$$\mathcal{S}(\mathcal{P}) = \{x : p_1 = \mathbf{s}, p_N = \mathbf{g}, \min_j d_{\mathcal{O}}(c_j(p)) \geq \rho \ \forall p \in \gamma(x)\}. \quad (1)$$

The objective is to learn a generative model $p_{\theta}(x \mid \mathcal{P})$ supported on $\mathcal{S}(\mathcal{P})$.

A rigid transformation $T \in \text{SE}(3)$ acts on a task by $T\mathcal{P} = (Ts, T\mathbf{g}, T\mathcal{O})$ and on a trajectory waypointwise, $(Tx)_i = Tp_i$. The task is then exactly $\text{SE}(3)$ -equivariant:

$$\mathcal{S}(T\mathcal{P}) = T\mathcal{S}(\mathcal{P}), \quad \forall T \in \text{SE}(3), \quad (2)$$

and a generative planner should inherit this,

$$p_{\theta}(Tx \mid T\mathcal{P}) = p_{\theta}(x \mid \mathcal{P}), \quad \forall T \in \text{SE}(3). \quad (3)$$

We use conditional flow matching [14], which learns a time-dependent velocity field $v_{\theta}(x_t, t, \mathcal{P})$ whose integral transports a simple prior to $p_{\theta}(x \mid \mathcal{P})$. Velocities are free vectors, therefore the corresponding condition on the field is

$$v_{\theta}(Tx_t, t, T\mathcal{P}) = R v_{\theta}(x_t, t, \mathcal{P}), \quad T = (R, \tau) \in \text{SE}(3), \quad (4)$$

which yields Eq. (3) whenever the prior is carried along with the task, $p_0(Tx \mid T\mathcal{P}) = p_0(x \mid \mathcal{P})$. No fixed prior does this, since no probability measure on $\mathcal{C}^N$ is translation-invariant, so the prior has to depend on $\mathcal{P}$. Nothing in the flow matching objective enforces either one, and Sec. V-E measures how far a trained field falls from Eq. (4).

B. Decomposing the group

The motion planning task gives us a canonical frame at initialisation, and at the cost of one cross product per query. Unfortunately the canonicalization is not entirely complete, since fixing the rotation about the first axis needs a second reference axis, and the query does not give us one. We decompose as follows.

We declare $m = \frac{1}{2}(\mathbf{s} + \mathbf{g})$, $\ell = \|\mathbf{g} - \mathbf{s}\| > 0$ and $\hat{u} = (\mathbf{g} - \mathbf{s})/\ell$. The second axis is not determined by the query, and ours is read off the world basis vector e_k that $\hat{u}$ is least aligned with,

$$k = \arg \min_i |\hat{u}_i|, \quad \hat{n} = \frac{\hat{u} \times e_k}{\|\hat{u} \times e_k\|}. \quad (5)$$

Choosing k this way bounds $\hat{u}_k^2 \leq \frac{1}{3}$, so $\|\hat{u} \times e_k\|^2 = 1 - \hat{u}_k^2 \geq \frac{2}{3}$ and the denominator stays away from zero for every query. Collecting the three axes into $R_{\mathcal{P}} \in \text{SO}(3)$, with $R_{\mathcal{P}}^{\top} = [\hat{u} \ \hat{n} \ \hat{u} \times \hat{n}]$, the reduction is the rigid map

$$\Phi_{\mathcal{P}}(p) = R_{\mathcal{P}}(p - m), \quad (6)$$

under which the endpoints are no longer inputs but constants,

$$\Phi_{\mathcal{P}}(\mathbf{s}) = (-\frac{\ell}{2}, 0, 0), \quad \Phi_{\mathcal{P}}(\mathbf{g}) = (+\frac{\ell}{2}, 0, 0). \quad (7)$$

Hence the network never sees a world coordinate. The task is carried into the frame, $\hat{x}_t = \Phi_{\mathcal{P}}(x_t)$ and $\hat{\mathcal{O}} = \Phi_{\mathcal{P}}\mathcal{O}$, obstacle centres transforming as points, box edges as free vectors under $R_{\mathcal{P}}$ alone, and radii not at all. The field is

evaluated entirely inside the frame, $\hat{v} = v_{\theta}(\hat{x}_t, t \mid \ell, \hat{\mathcal{O}})$, where by Eq. (7) the query enters as the single scalar ℓ in place of the six numbers of $(\mathbf{s}, \mathbf{g})$. The output is returned to the world by the rotation alone, $v = R_{\mathcal{P}}^{\top} \hat{v}$, velocities being free vectors. A prior drawn in the frame is carried with the task automatically, up to the roll of Prop. 1, which a prior isotropic about the first axis does not see.

Eq. (6) costs one cross product per query, constrains nothing about v_{θ} , and holds at initialisation. It fixes five of the six degrees of freedom of $\text{SE}(3)$, three of translation through m and two of rotation through $\hat{u}$. It does not fix the sixth, and Eq. (5) reads the world basis, so it turns with the world.

Proposition 1 (The residual is $\text{SO}(2)$). *Let $R_1(\theta)$ denote the rotation by θ about the first coordinate axis. For every task $\mathcal{P}$ with $\ell > 0$ and every $T \in \text{SE}(3)$ there is a $\theta = \theta(T, \mathcal{P})$ with*

$$\Phi_{T\mathcal{P}} \circ T = R_1(\theta) \circ \Phi_{\mathcal{P}}. \quad (8)$$

Proof. $\|T\mathbf{g} - Ts\| = \ell$, so by Eq. (7) both $\Phi_{T\mathcal{P}} \circ T$ and $\Phi_{\mathcal{P}}$ carry $\mathbf{s}$ to $(-\frac{\ell}{2}, 0, 0)$ and $\mathbf{g}$ to $(+\frac{\ell}{2}, 0, 0)$. Their composite $Q = \Phi_{T\mathcal{P}} \circ T \circ \Phi_{\mathcal{P}}^{-1}$ is, therefore, an orientation-preserving isometry fixing two distinct points, hence a rotation about the line joining them, which is the first coordinate axis. $\square$

Nothing in the proof refers to Eq. (5). The residual is $\text{SO}(2)$ for every rule selecting the second axis, and the rule decides only which θ . What Eq. (3) still asks of a planner in this frame is equivariance to $\text{SO}(2)$, then, and not to nothing. Escaping it would take a rule with $\theta(T, \mathcal{P}) = 0$ for every T , and Sec. III-C shows that no rule reading only the query has that property. Ours is in addition discontinuous: it jumps where the minimising k in Eq. (5) ties.

Eq. (6) can be read as input normalisation rather than as a group action, and the two readings name the same operation. The decomposition invites it: the three translations alone are worth +12.3 points, and recentring on m contains no rotation. What the symmetry reading adds is not a different computation but an account of which parts come off exactly, what is left when they do, and why the remainder is $\text{SO}(2)$ rather than a nuisance to be tuned away.

C. Why the query cannot remove the remainder

Sec. III-B removed five degrees of freedom by reading a frame off the query. Whether the sixth yields to the same treatment, under some better choice of second axis, is not a question about our construction. Three arguments answer it: one algebraic, one topological, and one that applies to an individual scene. None is original to us. We state them because the rest of the paper is an empirical answer to what they leave open.

Proposition 2 (The query fixes no roll). *Let $\Phi_{\mathcal{P}}$ be any rigid map that depends on $\mathcal{P}$ only through $(\mathbf{s}, \mathbf{g})$. Then $\Phi_{T\mathcal{P}} \circ T = \Phi_{\mathcal{P}}$ fails for every rotation $T \neq \text{id}$ about the line through $\mathbf{s}$ and $\mathbf{g}$.*

Proof. Such a T fixes $\mathbf{s}$ and $\mathbf{g}$, so $\Phi_{TP} = \Phi_P$ and the identity would require $\Phi_P \circ T = \Phi_P$. Since Φ_P is invertible, $T = \text{id}$. $\square$

No continuity was used, so the conclusion holds for discontinuous rules as well: the stabiliser of the query is exactly the residual of Prop. 1. Continuity fails separately.

Proposition 3 (Topological obstruction). *There is no continuous $\hat{n} : S^2 \rightarrow S^2$ with $\hat{n}(\hat{u}) \perp \hat{u}$ for every $\hat{u} \in S^2$. Every rule selecting the second axis from $\hat{u}$, Eq. (5) among them, is therefore discontinuous somewhere on the sphere of query directions.*

Proof. Such an $\hat{n}$ is a continuous nowhere-vanishing tangent vector field on S^2 , which the hairy ball theorem forbids [15]. $\square$

Proposition 4 (Scenes with their own symmetry). *Let $\hat{O}$ be a reduced scene with $R_1(\pi)\hat{O} = \hat{O}$. Then no rule reading only the reduced task can distinguish the frame R_P from $R_1(\pi)R_P$.*

Proof. $R_1(\pi)$ fixes both endpoints of Eq. (7) and fixes $\hat{O}$ by hypothesis, so the two frames present the rule with identical arguments. $\square$

A rule that also reads the scene escapes Prop. 2, and outside the measure-zero set of Prop. 4 it can fix the roll [12], though not continuously in general [6]. What we show is that the remainder cannot be removed the way the other five degrees of freedom were, in closed form from the query. Sec. V measures the three mechanisms that remain, and none of them recovers a point.

D. Three ways to impose the symmetry

A symmetry that the problem has and the model lacks can be imposed in three places. In the data, by training on randomly transformed copies of each problem, which teaches the symmetry without guaranteeing it. In the inference operator, by averaging the learned field over the group, which guarantees it for the averaged field without changing what was learned. In the weights, by restricting the hypothesis class to fields equivariant by construction, which guarantees it at initialisation and constrains optimisation throughout.

Averaging a field f over the K rolls $R_k = R_1(2\pi k/K)$, applied to the waypoints and the scene together, gives

$$(\mathcal{A}f)(\hat{x}, \hat{O}) = \frac{1}{K} \sum_{k=0}^{K-1} R_k^\top f(R_k \hat{x}, R_k \hat{O}), \quad (9)$$

with t and ℓ suppressed since the rolls leave them fixed. In $L^2(\mu)$, for any C_K -invariant measure μ on $(\hat{x}, \hat{O})$, $\mathcal{A}$ is the orthogonal projection of f onto the fields equivariant to the cyclic subgroup $C_K \subset \text{SO}(2)$ [18].

The first two mechanisms act on a field that has already been trained, and how far such a field sits from equivariant is measurable rather than assumed. We write

$$r = \frac{\|f - \mathcal{A}f\|}{\|\mathcal{A}f\|} \quad (10)$$

for the non-equivariance residual of f , with norms in that $L^2(\mu)$, which costs one forward pass per rotation and no retraining. It bounds what the second mechanism can do.

Proposition 5 (Symmetrisation budget). *Let f^* be C_K -equivariant and $\mathcal{A}$ the projection of Eq. (9). Then*

$$\|f - f^*\|^2 = \|\mathcal{A}f - f^*\|^2 + r^2 \|\mathcal{A}f\|^2. \quad (11)$$

The squared error that any equivariantisation of f can remove is at most $r^2 \|\mathcal{A}f\|^2$, and $\mathcal{A}$ attains it. The remainder $\|\mathcal{A}f - f^\|$ is equivariant error, which no symmetrisation of f reduces. Every map carrying f to an equivariant field moves it by at least $r\|f\|/\sqrt{1+r^2}$, with equality only for $\mathcal{A}$.*

Proof. $\mathcal{A}$ is an orthogonal projection, so $f - \mathcal{A}f$ is orthogonal to the equivariant subspace, which contains $\mathcal{A}f - f^*$. $\square$

IV. SETUP AND CALIBRATION

A. Benchmark and protocol

PointMass3D places forty obstacles, twenty spheres and twenty oriented boxes, in $[-1, 1]^3$. The robot is a sphere of radius $\rho = 0.03$. Collision checking uses an analytic signed distance field, and $\gamma(x)$ is resampled at a spacing of 0.01 before it is tested rather than evaluated at the 64 waypoints alone. Since d_O is 1-Lipschitz, every point of $\gamma(x)$ lies within 0.005 of a tested sample, so the reported rates differ from the continuous ones by at most a 0.005 perturbation of ρ , a sixth of the robot radius. A trajectory is $N = 64$ waypoints in $\mathbb{R}^3$.

Expert paths come from RRT-Connect, then random short-cutting, uniform resampling, and CHOMP refinement. The dataset is 300 environments, each with 600 start-goal pairs and 30 paths per pair including time reversals. Environments 250 to 299 are held out and never trained on. Unless stated otherwise, models train on the first 60 of the remaining 250; the scaling study in Sec. V-A varies that count.

The planner is a conditional flow-matching model with 2.16M parameters: a dilated temporal convolutional trunk [16] with FiLM conditioning [17] and a PointNet-style obstacle encoder [19]. The encoder max-pools all forty obstacles into a single 128-dimensional vector, and FiLM applies that vector identically to every one of the 64 waypoints.

Every arm transports the same prior, $\mathcal{N}(0, I)$ drawn in the frame that arm works in: world coordinates for the world-frame model, hence about the world origin, and the query midpoint m for the recentred and reduced models. Training draws x_0 the same way, after the reduction, so the prior the sampler starts from is the one the objective was fitted against. Endpoints are not imposed. The $p_1 = \mathbf{s}$, $p_N = \mathbf{g}$ of Eq. (1) is the target the model is fitted to, not a constraint applied at sampling time: the reported runs leave $\gamma(x)$ free at both ends and report the endpoint error separately, so the collision-free rates are not propped up by a constraint the model did not learn. Under the reduction the endpoints are the constants of Eq. (7) rather than inputs, which is consistent with its endpoint error being about a third of the world frame's, 0.0020 against 0.0059 at 250 environments. Both sit under a fifth of the robot

Planner	Free (%)	Clearance	s/query
straight line	15.6	0.055	<0.001
TrajOpt	74.0	0.026	0.352
CHOMP	88.8	0.019	0.394
RRT-Connect	99.6	0.017	0.284
expert (RRT+CHOMP)	100.0	0.037	0.342
flow, world frame	14.60	-0.071	0.053
flow, (s, g) reduction	51.10	-0.002	0.063

TABLE I

THE SAME 500 UNSEEN PROBLEMS SOLVED FIVE OTHER WAYS. THE LEARNED ROWS ARE CONVERGED, THREE-SEED MEANS AT 60 ENVIRONMENTS. CLASSICAL TIMINGS ARE ONE CPU CORE AND ONE QUERY; THE LEARNED ROWS ARE ONE GPU AND TWENTY SAMPLES, SO THE TIME COLUMN IS NOT LIKE-FOR-LIKE. CLEARANCE IS $d_O - \rho$, THE MARGIN BY WHICH THE ROBOT MISSES THE NEAREST OBSTACLE, MINIMISED ALONG THE DENSELY RESAMPLED PATH; IT IS NEGATIVE EXACTLY WHEN THE PATH COLLIDES. CLASSICAL ROWS AVERAGE IT OVER SOLVED PROBLEMS ONLY, LEARNED ROWS OVER ALL SAMPLES, WHICH IS WHY THE STRAIGHT LINE READS +0.055 WHILE FAILING 84.4% OF THEM. THE EXPERT ROW IS THE FILTERED TRAINING DISTRIBUTION RATHER THAN A PLANNER THAT NEVER FAILS.

radius $\rho = 0.03$, so neither rate is sensitive to how the ends are handled.

Every model is evaluated on the same 500 problems: 50 unseen environments, 10 distinct start-goal pairs each, 20 samples per pair, eight Euler steps. We report two metrics and never conflate them. A sample is *free* if the whole path is collision-free. *Best-of-20* asks whether any of the twenty samples for a query is free. At 60 environments they differ by 36 points. Uncertainty is quoted across seeds for claims about a method. For comparisons against a fixed baseline measured on the same problems we use a paired bootstrap over environments.

B. The floor, and a paired test against it

A learned planner is worth what it adds to the cheapest thing that solves the same problems. Joining start to goal with a straight segment is collision-free on 15.6% of the 500 held-out problems, so the clutter is dense and not so dense that the direct path is usually blocked. The world-frame model converges to $14.60 \pm 0.20\%$.

Quoting the across-seed spread here would answer a different question. That spread describes variation between training runs, not the uncertainty of a comparison against a fixed baseline measured on the identical problems. Both quantities track per-environment clutter, so we resample environments and recompute the difference within each replicate. The paired bootstrap gives -1.02 points, a 95% interval of $[-3.06, +0.90]$, and a standard error of 1.01, less than half the 2.2 obtained by resampling either rate alone, and still spanning zero. What we defend is therefore that training on world coordinates does not beat joining the endpoints with a line, and not that it loses to one. At 20 epochs even that was unavailable (15.37 ± 0.36). Optimisation is what makes the comparison resolvable, not what creates the deficit.

None of this is what the planner is worth to a practitioner. Behind a collision check, best-of-20, the reduced model solves 87.2% of held-out queries at 60 environments against 15.8%

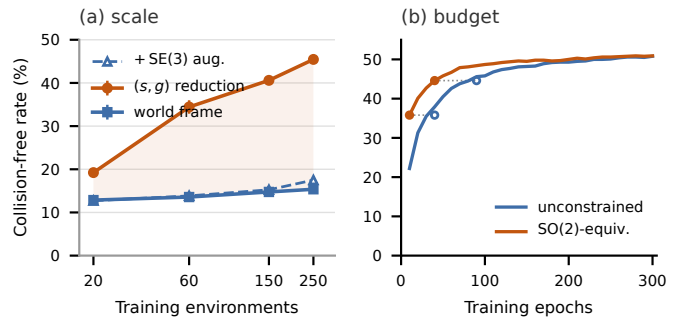


Fig. 2. (a) Held-out collision-free rate against training-set size, at a fixed budget of 20 epochs. The world-frame model moves 2.5 points across a $12.5\times$ increase in environments, from 12.9 to 15.4, while the reduction runs from 19.2 to 45.5 without plateauing. Holding the budget fixed is what makes representation the only variable, and it is also why these are not convergence levels: trained out, the 250-environment reduced model reaches 55.8. Every world-frame and reduction cell is a mean over three seeds. (b) The same rate against training epochs, 60 environments. The SO(2)-equivariant backbone reaches any given level two to three times sooner than the unconstrained model in the same frame, and the two converge to within 0.15 points. Seed-matched over three seeds.

for the best untrained sampler, a Brownian bridge about the straight segment at its best width. That is the number to read if the question is whether the sampler is useful rather than what the representation buys, subject to Table I.

V. RESULTS

A. Scaling

Across a $12.5\times$ increase in training environments the world-frame model goes from 12.9 to 15.4. The reduction goes from 19.2 to 45.5, and has not plateaued. On this grid the frame is worth +30.1 points, splitting into +12.3 for the three translations and +17.8 for the two rotations. The +12.3 includes prior placement: the world-frame arm draws the flow-matching prior as $\mathcal{N}(0, I)$ about the world origin, while both recentred arms draw it about the query midpoint m , so recentring moves the prior onto the query at the same time as it moves the target. An isotropic prior is unchanged by a rotation, so the +17.8 carries no such term. Seed-matched, the difference grows monotonically at every size: $+6.37 \pm 0.17$, $+20.87 \pm 1.54$, $+25.85 \pm 0.51$ and $+30.08 \pm 0.44$, quoting the standard error of the per-seed differences rather than combining the two arms' spreads in quadrature.

The world-frame model is not short of data. Training it on random SE(3) motions of the whole problem, which is the data mechanism of Sec. III-D, moves it by at most 2.1 points and leaves it at 17.5. These cells hold the budget at 20 epochs so that representation is the only thing varying, which is why 45.5 at 250 environments sits below the 51.10 that Table I reports at 60. Trained out, the 250-environment model reaches 55.8.

B. Control: a query-conditioned encoder

The gap admits a deflationary reading. In the world frame the obstacle encoder sees obstacles and nothing else, so its output is constant across every query in an environment, by

construction rather than by training. Under the reduction the same encoder sees those obstacles expressed in the query’s own frame, so its code varies with the query. On this reading the +36.5 measures letting the encoder see the query, and the change of frame is incidental.

We built the model that separates the two. It stays in the world frame throughout, and the raw query is concatenated to every obstacle before the pool, so the scene code depends on the query with no change of frame anywhere. It carries 0.07% more parameters.

It converges to $15.59 \pm 0.38\%$ against the world frame’s 14.60 ± 0.20 , seed-matched +1.01 and positive on all three seeds. Query dependence is worth about one point of the 36.5. The level it reaches is the more useful number. At 15.59 it lands on the straight line’s 15.6, so showing the encoder the query is enough to reach the floor and not enough to pass it.

C. Control: local geometry, and how it scales

One scene code applied identically to all 64 waypoints leaves a waypoint no channel through which to ask what is near *it*, and collision avoidance is that question. Sec. V-B shows the reduction addresses this indirectly. It can also be addressed directly. We append to each waypoint the signed distance to the scene and its gradient there, which for the $SO(2)$ -equivariant backbone splits by irreducible representation: d_O and the axial component of ∇d_O are invariant, the normal component rotates. This is a featurisation and not an oracle, since the signed distance is a deterministic function of primitives the network already receives. What it assumes is that those primitives are available at inference, which holds on this benchmark and fails for a planner operating on perception.

Table II crosses the two. Local geometry is worth +40.0 points in the world frame, more than the frame itself is worth, so the representation change studied here is not the largest effect available on this benchmark. The two are not substitutes either: with local geometry supplied, the frame still adds +24.8. Adding the two effects predicts $14.60 + 36.5 + 40.0 = 91.1$, and the measured value is 79.37. Whether that is an interaction depends on the scale. Adding on the logit scale predicts 88.0. If each intervention removed an independent fraction of failures, with q the failure rates, the prediction would be $1 - q_F q_L / q_0 = 74.0\%$, which the measurement exceeds.

What the frame is worth beside local geometry decays with data. Differencing within seed at a matched 80-epoch budget, it is $+22.37 \pm 0.46$ points at 60 environments and $+17.24 \pm 0.17$ at 250, two seeds at each scale. With two seeds each standard error rests on a single difference, so the comparison is made per seed, and on either seed the decay exceeds 4 points. Local geometry is what scales over that range, gaining 11 points while the frame beside it loses 5.

D. Replications

Three things vary, one at a time. A second state space: an L-shaped rigid body in the same clutter, whose state is a full pose rather than a point, so the points-versus-vectors distinction

60 envs, converged	global only	+ local	geometry adds
world frame	14.60	54.54	+40.0
(s, g) reduction	51.10	79.37	+28.3
+ equivariant backbone	51.27	82.13	+30.9
<i>the frame contributes</i>	+36.5	+24.8	

TABLE II

LOCAL GEOMETRY AGAINST THE QUERY FRAME. EVERY CELL IS A MEAN OVER TWO OR THREE SEEDS. THE TWO LOCAL-GEOMETRY COLUMNS ARE THE LEAST SEED-SENSITIVE MEASUREMENTS IN THIS WORK, WITH ACROSS-SEED STANDARD ERRORS OF 0.01 AND 0.10 AGAINST 0.20 AND 0.25 WITHOUT. THE EQUIVARIANT ROW IS A SEED-MATCHED PAIR, 51.27 AGAINST 51.12 ON THE SAME THREE SEEDS, SO WHAT THAT BACKBONE IS WORTH IS THE +0.15 BETWEEN THEM AND NOT 51.27 MINUS THE THREE-SEED MEAN 51.10 ABOVE IT.

Replication	Condition	World	Reduced	Gap
Second state space. SE(3) rigid body, floor 4.8				
	20 ep.	2.8	6.2	+3.4
	155/159 ep.	3.1	10.0	+6.9
Other objective. DDPM, matched network evaluations				
	20 ep.	12.27	28.97	+16.71
	300 ep.	13.51	50.52	+37.01

TABLE III

THE EFFECT SURVIVES A CHANGE OF STATE SPACE, AND A CHANGE OF GENERATIVE OBJECTIVE. THE RIGID BODY IS AN L-SHAPED UNION OF FIVE SPHERES WHOSE STATE IS A FULL POSE, CHOSEN TO HAVE TRIVIAL SYMMETRY SO THE DOMAIN IS NOT SECRETLY EASIER; THE WORLD-FRAME MODEL STAYS BELOW ITS FLOOR AT BOTH BUDGETS AND THE REDUCED ONE CLEARS IT AT BOTH. THE SECOND RIGID-BODY ROW IS EACH MODEL’S OWN BEST-VALIDATION CHECKPOINT, EPOCH 155 FOR THE WORLD FRAME AND 159 FOR THE REDUCTION, SO NEITHER HAD STOPPED IMPROVING WHEN THE RUNS ENDED. SINGLE SEED EXCEPT THE DDPM 20-EPOCH ROW.

moves inside the trajectory. Its trivial baseline is geodesic interpolation, straight in position and slerp in rotation. The world-frame model never clears that floor at either budget; the reduction clears it and is still climbing. A second generative objective: replacing flow matching with DDPM at matched network evaluations leaves the gap at +37.01 against +36.50, so the effect does not belong to the sampler. A range of clutter: on freshly generated layouts at densities neither model was trained on, the gap runs from +29.4 at eight obstacles to +41.2 at twenty-four.

The rigid-body cells are the weakest measurements in this paper: one seed, and a world-frame model below its floor at both budgets. What replicates there is the ratio and not the level. At the best-validation checkpoints the reduced model is $3.2\times$ the world-frame model, against $3.5\times$ for the converged point mass.

E. The residual, and what it bounds

Proposition 5 bounds what symmetrising a trained field can remove. The reduced model has $r = 0.017$, so frame averaging can remove under 3×10^{-4} of the field’s squared magnitude. The bound predicts that averaging changes the field little, and the measured +0.68 agrees. It does not convert a field error into a collision rate, which a small change near an obstacle can flip. The measurement costs one forward pass per rotation and no retraining.

The residual does not forecast what a mechanism is worth. Eq. (10) averages over the roll, which the world-frame model does not have, so to compare the two arms we average instead over the group the problem has. For each held-out query we draw K rotations Q_k from the Haar measure on $SO(3)$, apply each to the whole problem (endpoints, obstacle centres, box edges and the state), evaluate the field in the transformed copy, and carry the output back by $Q_k^\top$, a velocity taking the rotation alone. For the reduced arm the reduction is applied inside each copy, so what is measured is the residual of the pipeline rather than of the network, and the probe state is drawn where each arm’s own sampler draws it. Translations are held at zero, since a translated scene also leaves the training distribution and would confound non-equivariance with extrapolation. Rotations are not innocent on that count either, since rotating $[-1, 1]^3$ about the origin carries its corners out to radius $\sqrt{3}$, so the world frame’s 0.0881 mixes the two as well; the comparison we lean on is unaffected, because the augmented arm trained on exactly these motions.

Here K is a Monte-Carlo sample size, not the cyclic order of Eq. (9): the Q_k are random rather than a subgroup, so $\mathcal{A}$ is an estimate of the Haar average rather than an exact projection, and normalising the spread by K rather than $K - 1$ biases r low by $\sqrt{(K - 1)/K}$. Correcting that factor removes most of the drift of r with K and leaves a scatter of about 3%. Every value quoted here is measured at $K = 8$ and left uncorrected, where the factor is 0.935, so a corrected reading differs by 6.5%. Augmentation drives r from 0.0881 to 0.0181 and buys 2.1 points; the reduction, at 0.0174, buys 30.1. The two residuals differ by about 4%, comparable to that scatter, and both are single measurements on one checkpoint each, since r is a property of a particular trained field.

VI. LIMITATIONS

- **The group must act on the state space, not just the workspace.** This holds for a point mass and for the rigid body of Table III. It fails for a manipulator planning in joint space: the query still defines a frame in the workspace, but a rigid motion of the world induces no clean action on joint angles. Mobile bases, free-flying bodies and end-effector-space planning satisfy the condition. Joint-space planning for arms does not.
- **The benchmark was chosen for isolability, not for difficulty.** RRT-Connect solves 99.6% of these problems in under 0.3 s on one CPU core, and on this benchmark one should use RRT-Connect. A point mass in a box is where clutter, budget, seed and representation can be varied one at a time, which is the only reason any number here is attributable to the representation. The sampler is not a toy at that: behind a collision check it solves 87.2% of held-out queries.
- **One workspace family, one clutter generator, one architecture.** The claim that the sixth degree of freedom needs no treatment relies on the training distribution supplying implicit roll diversity through many start-goal

directions. A dataset with few directions per environment could reverse it. The architecture also pools the scene into one code applied to every waypoint. A trunk in which each waypoint attends to the obstacles widens that path, and Sec. V-C suggests the frame would be worth less there.

- **One cell of the design was not run.** The $SO(2)$ -equivariant backbone was only ever trained inside the reduced frame, so what we establish is that architectural equivariance adds nothing once the frame is present. That is not the same as showing it cannot supply what the frame supplies. A world-frame equivariant model would separate the two readings, and it is the experiment we would run next.
- **Seeds are not uniform.** The 60-environment cells carry three seeds, the local-geometry cells two or three, and the 250-environment grid three. Only the converged 250-environment run and the rigid-body cells of Table III carry one. Every claim we lean on has at least two.
- **The frame’s value beside local geometry falls with data.** It is +22.4 points at 60 environments and +17.2 at 250 on a matched budget. We have not found where that decay stops, and a reader extrapolating to much larger datasets should expect the margin to narrow rather than hold.
- **Local geometry assumes explicit primitives at inference.** The signed distance is a deterministic function of obstacle data the network already receives, so it is a featurisation rather than an oracle, but it is unavailable to a planner working from perception.

VII. CONCLUSION

Five of the six degrees of freedom of $SE(3)$ were removable exactly, at initialisation and at the cost of one cross product per query, by a frame the planning query already supplied. That was worth +36.5 points of held-out collision-free rate, against a world-frame model that did not beat a straight line. The sixth was not removable from the query by any rule. Frame averaging and the $SO(2)$ -equivariant backbone were trained to convergence, and neither recovered a point. Roll augmentation was measured only at a fixed budget, where its +0.8 already sat inside the spread across three seeds. The $SO(2)$ -equivariant backbone still earned its place, reaching any given level two to three times sooner, so the symmetry bought optimisation and not accuracy. Two models with indistinguishable residuals differed by 28 points, so the residual was not sufficient to predict what a mechanism was worth. Giving each waypoint its distance and direction to the nearest obstacle was worth more than the frame was, and with that information supplied the frame still added +24.8 points.

The construction transfers wherever the group acts on the state space and the query names two points, which covers mobile bases, free-flying bodies and end-effector-space planning, and excludes joint-space planning for arms. It costs one cross product per query and constrains nothing about the network.

The payoff should be largest where relearning the same motion at every pose is most expensive, which means a model small relative to its workspace and a narrow conditioning path from the scene to the trajectory. Both were true here, and the +36.5 points shrank to +24.8 as soon as we widened that path with local geometry.

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The software implementation was carried out with Claude Code under the author’s direction. The classical planner implementations, used to generate expert trajectories and to bound what is achievable, follow the published algorithms [13], [22], [27] and their public reference implementations. The problem formulation and the experimental design are the author’s own, as is the theoretical analysis except as noted here. Claude was used for language and grammar editing of the manuscript, and proposed corrections to the analysis and to reported numbers, including the argument of Prop. 2. The paragraph of Sec. IV-A stating the prior and the endpoint convention, and the second paragraph of Sec. V-E stating how the residual is measured under rotations, were drafted with Claude Code from the released code and checked against it by the author. Every reported number is produced by the released code, whose geometric and equivariance-critical components are verified by property-based tests that run on untrained networks and include negative controls.

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